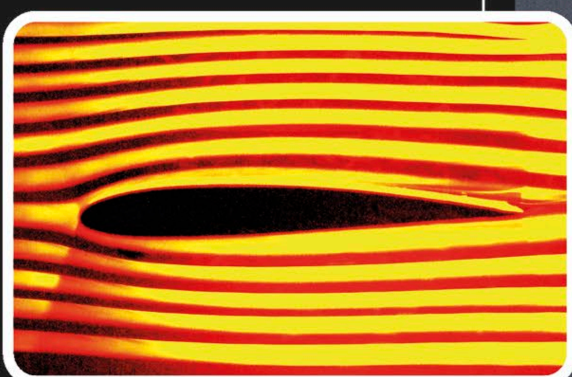
SAVING SECONDS

WIND TUNNELS

A moving bicycle pushes air out of the way as it moves forward, but this becomes harder to do as its speed increases. The study of how air flows around an object, called aerodynamics, helps scientists find ways of making vehicles more streamlined so that they can pass through air more easily and quickly. They use wind tunnels to test the aerodynamics of bicycles, cars, aircraft, spacecraft, and sports equipment. In a wind tunnel, air is blown at vehicles through powerful fans and smoke is added to see how the air flows around it.

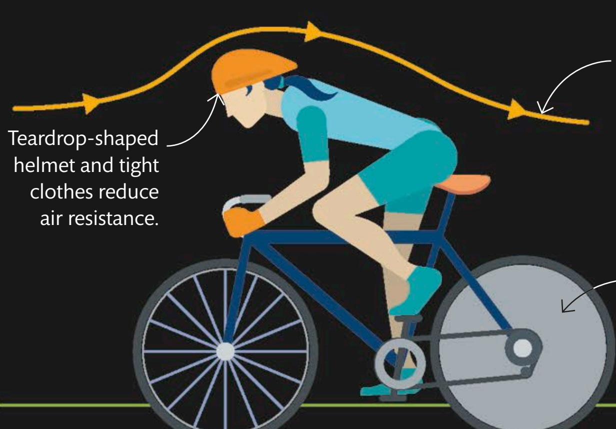
AIRCRAFT WINGS

Airplane designers use aerodynamics to design wings so that air flows faster above the wing than below it. That means that the air pressure is higher beneath the wing, resulting in an upward force, called lift, which enables aircraft to take off and remain airborne.



AIR RESISTANCE

An object moving through air experiences air resistance, also known as drag. This is a force that acts against the object's direction of movement. Minimizing this resistance makes it easier for the object to speed up, or accelerate. When a cyclist leans forward, the curved shape of their body allows air to flow around them more easily. This reduces air resistance, making it easier for the cyclist to accelerate.



Built by NASA, the largest wind tunnel in the world is more than 1,400 ft (427 m) long.